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ACRONYMS

ASI	Autonomous Solutions Inc.
EKF	extended Kalman filter
FOG	fiber optic gyro
GPS	global positioning system
LOS	line of sight
NLOS	non-line of sight
RTK	real-time kinematic
TOF	time of flight
UGV	unmanned ground vehicle
UWB	ultra wideband radios

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CHAPTER 1

INTRODUCTION

Autonomous convoy operations can fill growing needs within military, mining, agriculture, and industrial fields. The problem however, is each of these fields have situations in which an autonomous convoy solution is inoperable because no GPS position can be acquired. GPS denied environments cause each unmanned ground vehicle (UGV) in the convoy to loose positional accuracy, and can lead to inoperability, or even damage.

Research has been done using ultra wideband (UWB) radios as a solution to the GPS denied problem. [1, 2] However, these research groups only performed analyses of a UWB positioning system, and not the performance of a system which depends on UWB measurements. In order for autonomous convoy systems to fill the needs of any particular field, a well performing system must be designed such that a convoy can operate in GPS denied environments.

The main objective of this research is to use UWB radios to localize a leader vehicle within a convoy situation while in a GPS denied environment. The position of the leader should be within one meter of accuracy, speeds up to 10 m/s, and distance of 50m between vehicles. The secondary objective of this research is to augment the UWB localization algorithm with GPS measurements when available.

This research will be done in partnership with Autonomous Solutions Inc (ASI). ASI will be providing:

- five Time Domain P440 UWB radios
- access to two Ford Escapes each outfitted with a full automation kit
- access to closed course test tracks
- advice and guidance from the ASI Research Team

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Their support provides an existing platform to design, test, and build a working GPS denied leader localization system.

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CHAPTER 2

Review of Literature

2.1 Motivation

GPS denied environments are ubiquitous throughout the military, mining, agriculture and industrial fields. Examples of GPS denied environments include inside buildings, factories, mines,dense forests, canyons, and even when the GNSS signal is being jammed by opposing military forces. As these situations are common, finding a method of localization is essential to push the technologies into widespread use.

2.2 Kalman Filter

Nearly all the articles reviewed use a type or extension of the popular Kalman filter [2–4]. One requirement for the Kalman filter to provide optimal state estimates is the noise distribution on the sensor measurements must be Gaussian. [5, pg 151] Both Richardson and Shen show that the range noise distribution is Gaussian for the UWB ranges. [1, 3] Therefore the Kalman filter will give optimal estimates from the UWB measurements.

Different types of Kalman filters have been used by each research group. Richardson used a modified extended Kalman filter and a modified regularized particle filter. Shen, Vydhyanathan, and Guo used an extended Kalman filter. [2–4] Guo mentions that their EKF was initialized using a trilateration algorithm similar to a similar geometric solver.

Non-probabilistic solvers have also been used by a few sources. Ruiz suggests a least-squares method for geometric bilateration using circle intersections. [6] Richardson used a simple law of cosines based solver for their bilateration. [1]

2.3 Radio Configuration

Two different UWB radio placement configurations have been shown in the literature reviewed:

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- Three radios, one mobile and two anchors
- Four radios, one mobile and three anchors

where anchor radios are assumed to be at fixed, known locations and mobile radio positions are to be estimated. These articles show either three or four radio configurations will work for localization. [1–4]

The primary purpose of using more than three radios is to remove ambiguity from the geometric solution. [1] Other articles discount or ignore this problem by using a Kalman filter to resolve which solution is correct.

2.4 GPS Blending

Shen and Vydhyanathan used GPS alongside the UWB measurements in order to blend the two sensors together and increase accuracy of the result. [2, 3] GPS/UWB sensor blending is another reason that the research groups opted to use a Kalman filter.

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CHAPTER 3

Research and Design Methods

3.1 Sensor Characteristic Identification

Before work begins on localization algorithm development, an understanding of how the UWB sensor will operate within the track environment at ASI. The Time Domain PulseOn P440 ultra wideband radio, pictured in figure 3.1, will be used for the ranging measurements. A characterization of the P440 radio will need to include:

- accuracy and variance measurements
- sample rates with various radio settings
- sample rates at ranges 0-100 meters
- optimal radio settings for convoy operations

Radio characteristics will need to be made in order to properly model the radio behavior within a simulation environment.

3.2 Simulation Development

Simulation provides an excellent environment in which to develop localization algorithms. The simulation environment for this research needs to represent at least two vehicles, and characterize the P440 UWB radio. The goal is to develop vehicle and radio models which will behave as closely to the real dynamics as possible. With accurate models, localization algorithms can be developed with confidence that they will work the same way on the physical hardware.

The simulation environment will also reduce the localization algorithm development time and increase safety while testing.

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Fig. 3.1: The Time Domain P440 ultra wideband radio.

3.3 Integration

As stated in the introduction, ASI is providing an existing autonomous convoy vehicle platform upon which this research will build. The existing convoy platform is capable of leader-follower type missions using GPS for localization. The following steps will need to be taken to evaluate the UWB localization on the test vehicles:

1. Mount P440 Radios on the vehicles and wire them accordingly.
2. Patch the existing convoy software to use the localization algorithm developed in simulation.
3. Test the patched software in the convoy simulation environment.
4. Test the patched software on the test track at ASI.

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3.4 Accuracy and Usability Study

After the localization algorithm is running on the two vehicles, data will be collected in order to validate the position estimates are within specifications. GPS data will be used as a baseline against which the position estimate will be compared. The comparison will show:

- Cross-track error
- Down-track error
- heading error

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CHAPTER 4

Conclusion

GPS denied localization systems must be developed in order to make autonomous convoy systems feasible and useful. Ultra wideband radios are the proposed solution to enable autonomous convoys to operated in any environment. These convoy systems have the potential to lower running costs, reduce man-power, and increase safety in many fields including military, agriculture, mining, and industrial. [7]

By simulating UWB radio measurements, radio characteristics, and vehicle dynamics, an algorithm can be developed to localize a leader vehicle without the used of GPS. Then simulating the UWB localization algorithm in a convoy simulator, the algorithm can be tested to work before integrating into a working convoy system.

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